

AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

Project Proposal Report

R26-IT-050

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B.Sc. (Hons) Degree Information Technology Specializing in Information Technology

Department of Information Technology

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of Supervisor Date

Signature of Co-Supervisor Date

Abstract

Tourism in Sri Lanka is growing fast, but travelers still face many problems when they go to areas with difficult roads like Ella or Nuwara Eliya. Most people use apps like Google Maps, but these apps only show the shortest way. They don't check if the vehicle traveler chose is safe for that specific road and weather. For example, a small 800cc car might have engine problems on very steep roads when carrying many people.

This research focuses on making an intelligent engine for vehicle selection and route optimization. I am using a new technology called **Graph Retrieval Augmented Generation (GraphRAG)** and **Large Language Models (LLM)**. My system asks the traveler for details like where they go, how many people, and their budget. Then, the system checks real time data from Weather APIs, Google Elevation data, and Disaster Management Centre (DMC) alerts.

The main thing in this research is to see the connection between the vehicle's engine capacity (CC) and the road steepness (gradient). If the traveler's requested vehicle is not safe for that road or weather, the AI model will suggest a better vehicle. Also, the system suggests a "scenic route" so tourists can see beautiful places (POIs) and it calculates the "True Trip Cost" based on the engine power and road difficulty.

By using GraphRAG, the system can give very accurate and safe suggestions. This helps to reduce accidents and give a better travel experience for tourists in Sri Lanka.

Keywords: GraphRAG, Vehicle Selection, Route Optimization, Sri Lankan Tourism, Context Aware AI.

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1. Introduction

Tourism is one of the most important things for Sri Lanka's economy and many people love to travel to places like Nuwara Eliya and Ella. These areas have very beautiful mountains but the roads are very steep and difficult to drive. When tourists come to Sri Lanka, they usually book a vehicle but they don't know if that vehicle is safe for these mountain roads. Currently, many navigation apps like Google Maps show only the shortest way but they don't think about the "Context" of the travel. For example, if there is heavy rain, riding a bike on a mountain road is not safe. Also, small cars with low engine capacity will struggle to climb steep hills when there are many passengers. This can cause engine problems or even accidents. According to Mohamed et al. (2026) [1], using weather aware data is very important for safe routing in modern systems. In Sri Lanka, road elevation is also a big factor because it changes how much fuel is used and how safe the journey is.

There are some big problems in current travel apps because they only look at the distance and they don't check if the vehicle's engine capacity (CC) is enough for the road's gradient (steepness). Even though Sri Lanka has a Disaster Management Centre (DMC), their alerts about landslides are not connected to the travel apps. This makes the "True Trip Cost" often wrong because driving on kanda parawal uses more fuel than normal roads. Also, standard AI models like ChatGPT can give travel suggestions but sometimes they give wrong info because they don't have a structured "Knowledge Graph" of Sri Lankan roads. Recent studies by Naganawa et al. (2025) [2] show that GraphRAG is a better technology to solve these kinds of problems by connecting different data parts together.

To solve all these issues, I am building an Intelligent Multi Criteria Vehicle Selection and Route Optimization Engine. My system will use GraphRAG and Large Language Models (LLM) to give better suggestions. The traveler will give their destination, budget, and vehicle preference, and the system will check real time weather and elevation data. If the vehicle is not safe for that road, the system will suggest a better one. It will also show a "Scenic Route" with verified points of interest while calculating the correct fuel cost based on the vehicle's engine and road difficulty. This way, tourists can have a safe and beautiful travel experience in Sri Lanka.

1.1 Background and Literature Survey

Tourism serves as a critical pillar for Sri Lanka's economy, with high traveler interest in exploring mountainous regions such as Nuwara Eliya and Ella. However, the steep road gradients and unpredictable weather in these areas present significant safety challenges for drivers [10]. Existing navigation tools like Google Maps primarily focus on distance and travel time, often ignoring the mechanical limitations and contextual factors of the vehicle being used [9]. For instance, a vehicle with low engine capacity (CC) may experience severe engine strain or failure when attempting to climb a 20% gradient road under a full passenger load. Research has confirmed that accurately identifying both vehicle mass and road gradient is essential for maintaining engine control and optimizing fuel efficiency [11]. Furthermore, the gradeability of a vehicle its ability to climb slopes is directly impacted by the road's steepness and the vehicle's gear performance [10].

Integrating real time environmental data is paramount for safe routing in these dynamic terrains. Recent studies emphasize that weather aware routing using transformer based models is crucial for preventing accidents in high risk zones [1]. Additionally, experimental studies have shown that analyzing driving modes and engine parameters provides deeper insights into how terrain impacts overall vehicle load and emissions [13]. Beyond safety, the spatial layout and accessibility of rural tourism destinations play a significant role in enhancing the overall traveler experience [14]. While previous research utilized Multi Criteria Decision Making (MCDM) techniques for hotel selection, the same logic can be applied to evaluate vehicle suitability based on factors such as budget, passenger count, and mechanical power [6]. Similarly, safety frameworks have successfully applied spatial clustering techniques to identify and avoid high risk areas during route planning [5]. This approach can be extended by integrating real time disaster alerts such as landslide and flood warnings from the Disaster Management Centre (DMC) of Sri Lanka [8].

A major limitation of conventional AI models is their lack of structured and localized knowledge, which often leads to inaccurate or generic travel suggestions. The emergence of GraphRAG technology offers a promising solution by integrating Knowledge Graphs with Large Language Models to enable structured and context aware information retrieval [2]. Deep GraphRAG architectures further enhance this capability by identifying hierarchical relationships within the data, significantly reducing hallucinations and improving response accuracy [12]. The effectiveness of Retrieval Augmented Generation (RAG) has also been demonstrated in complex transportation systems, where hybrid RAG models can combine multiple data sources for more reliable navigation and decision making [4]. By bridging the gap between vehicle engine capacity, road steepness, and real time environmental conditions, this research aims to develop a safer and more context aware travel support system for tourists traveling in Sri Lanka.

1.2 Research Gap

A critical analysis of the existing literature and current navigation systems reveals several unresolved gaps in the tourism and mobility sector:

- **Absence of Vehicle to Terrain Matching:** While research has established that vehicle mass and road gradients are essential for performance, current systems like Google Maps do not correlate specific engine capacities (CC) or torque with real time road gradients to determine mechanicalsafety.

- **Disconnect in Real time Safety Integration:** Although frameworks exist for safe routing based on crime data, there is a lack of integration regarding localized, high risk environmental hazards such as landslide and flood alerts provided by the Disaster Management Centre (DMC) of Sri Lanka.

- **Inaccurate Cost Estimation Models:** Existing platforms calculate travel costs based solely on distance. However, studies show that terrain difficulty and driving modes significantly impact fuel consumption, yet no system currently calculates a "True Trip Cost" by fusing engine load data with road steepness.

- **Underutilization of Structured Reasoning:** While GraphRAG technology has been proven to enhance information reliability and reduce AI hallucinations, it has not been applied within the Sri Lankan tourism context to balance a traveler’s financial budget with a vehicle's mechanical feasibility and safety requirements.

Table 1: Comparison of the Technologies/Methods in Existing Research vs Proposed Method

No table of figures entries found.	**Mohamed (2026)**	**Liu (2024)**	**Jagtap (2026)**	**Kollu (2025)**	**Naganawa (2025)**	**Zhang (2025)**	**Proposed Method**
Focus on Vehicle to Terrain Matching (Engine CC vs Gradient)							✓
Real time API Data Fusion (Weather/DMC Alerts)	✓						✓
Use of GraphRAG for Semantic Reasoning		✓			✓		✓
MultiC riteria Decision Making (Budget + Safety + Preference)			✓			✓	
Localized Safety First Dynamic Rerouting (SL Context)				✓		✓	
Explainable AI (XAI) for Travel Alerts					✓	✓	
Integration of Multi factors (Engine CC + Weather + POIs)							✓

1.3 Research Problem

The main problem this research wants to solve is how to give a safe and accurate vehicle recommendation for tourists in Sri Lanka by looking at many difficult factors at the same time. Even though there are many navigation tools, they have a big technical problem because they don't understand the relationship between "Vehicle Mechanics" and "Environmental Conditions." For example, current systems cannot reason if a specific car with a certain engine CC is powerful enough to go on a very steep road when the weather is bad. This is a complex problem because it needs to connect different types of data like vehicle specs, live weather, and road elevation which are usually not connected.

Another part of the research problem is that standard AI models like LLMs often make mistakes (hallucinations) about local road safety because they don't have a structured "Knowledge Base" for Sri Lankan context. Without a proper way to link a tourist's budget with the actual "True Trip Cost" (which changes based on the road difficulty), travelers often face unexpected costs or safety risks. Therefore, the core research problem is to build a reasoning engine using **GraphRAG** that can balance the traveler's preference, the vehicle's engine capacity, and real time safety alerts to suggest the most suitable and safest travel plan.

1.4 Research Questions

In this research, I am trying to find answers for these main questions to solve the problems in the tourism sector:

1. How can a **GraphRAG based reasoning engine** connect different data like vehicle engine capacity (CC), road steepness (gradient), and real time weather to give a safe recommendation?

2. What is the best way to integrate real time alerts from the **Disaster Management Centre (DMC)** and weather APIs into a travel planning system to avoid risky areas like landslides?

3. How does the connection between the vehicle's engine power and the road's elevation affect the **"True Trip Cost"**(fuel consumption) compared to normal distance based calculations?

4. How can Large Language Models (LLMs) provide more accurate and "hallucination free" travel suggestions by using a structured **Knowledge Graph** of Sri Lankan roads and tourist spots?

5. How can the system balance the traveler's personal budget and their vehicle preference with the necessary safety rules required for difficult terrains in Sri Lanka?

2. OBJECTIVES

2.1 Main Objective

The primary objective of this research is to develop an intelligent vehicle selection and route optimization engine utilizing **GraphRAG** and **Large Language Models (LLM)** . The system aims to provide tourists in Sri Lanka with the safest vehicle recommendations and optimized travel paths by analyzing the critical correlations between vehicle engine capacity (CC), road gradients, real time weather conditions, and the traveler's financial budget.

2.2 Sub Objectives

To achieve the main objective within a twelve month duration, the following sub objectives will be executed:

- **To design and implement a Tourism Knowledge Graph:** Develop a structured data model using Neo4j that integrates road types, elevation data, and vehicle mechanical specifications (CC and Torque) to facilitate AI driven reasoning.
- **To develop a Vehicle to Terrain Matching Algorithm:** Create a logic engine that assesses the mechanical feasibility of a vehicle for a specific route by comparing engine power against road steepness and passenger load.
- **To integrate Real time Environmental and Hazard Data:** Connect the system to external APIs, including Weather services and the Disaster Management Centre (DMC), to enable dynamic path planning based on live landslide and flood alerts.
- **To engineer a Scenic Route Optimizer:** Suggest alternative travel paths that highlight verified points of interest (POIs) while maintaining strict adherence to safety standards and the user's budget constraints.
- **To formulate a Terrain Aware Costing Model:** Establish a mathematical formula to calculate accurate fuel costs by factoring in engine strain on high gradient roads rather than relying on standard distance based metrics.
- **To validate System Accuracy and Performance:** Evaluate the reliability of the GraphRAG. based suggestions through real world testing scenarios and comparative analysis with traditional navigation applications.

3. METHODOLOGY

In this section, I am explaining how I build the Intelligent Vehicle Selection and Route Optimization system step by step. The main technology I use is **GraphRAG** because it can connect different types of data like road maps and vehicle engine specs to give safe travel advice for Sri Lankan tourists. First, I start with "Data Collection" where I get information about roads in Sri Lanka, their steepness (elevation), and different vehicle types with their engine capacity (CC). I am not using a normal database, but I am building a **Knowledge Graph** using Neo4j. This graph helps the AI to understand that a small 800cc car is not safe for a road with a 20% gradient when it is full of people.

The next part of the methodology is the "Reasoning Engine" where I connect a Large Language Model (LLM) with my Knowledge Graph. When a traveler enters their budget and destination, the system retrieves specific facts from the graph about the road condition and the vehicle's mechanical limits. I developed a special "Vehicle to Terrain Matching Logic" to calculate a safety score. This logic checks three main things: the Engine CC, the total weight of passengers, and the road steepness. If the weather is bad (like heavy rain), the system will use real time data from Weather APIs to update the safety score. This is very important for bikes and small cars in mountain areas like Ella or Nuwara Eliya.

Finally, the system does "Route Optimization" to suggest the best path for the tourist. If the shortest path is dangerous because of a landslide alert from the Disaster Management Centre (DMC), the system will find an alternative "Scenic Route". This route is not only safe but also shows beautiful points of interest (POIs) that match the traveler's interest. The system also calculates the "True Trip Cost" by looking at the engine strain on hills, so the traveler's budget is always accurate. To evaluate the system, I will compare the results with Google Maps and ask some travel experts to check if the safety warnings are correct and helpful for tourists.

3.1 Tools and Technologies

To build the Intelligent Vehicle Selection and Route Optimization system, I am using the following tools and technologies:

Backend & API Management:

- **Node.js:** Serving as the main backend runtime to handle real time data from travelers and manage the safety reasoning logic.
- **Express.js:** Used for creating RESTful APIs to connect the AI reasoning engine with the mobile application.

AI & Reasoning Core:

- **Gemini 1.5 Pro API:** Utilized for high level semantic reasoning to understand the traveler's context and generate justifications for vehicle recommendations.
- **LangChain:** Used as the orchestration framework to link the LLM with the Knowledge Graph and external data sources.

Graph & Knowledge Base:

- **Neo4j (Graph Database):** Used to build the Knowledge Graph that connects vehicle specs (Engine CC), road gradients, and safety rules for accurate reasoning.
- **MongoDB Atlas:** For storing user profiles, past travel logs, and unstructured metadata related to points of interest (POIs).

Frontend & Mobile Application:

- **React Native:** For building the cross platform mobile app so tourists can get real time vehicle suggestions and navigation alerts on both Android and iOS.
- **Next.js:** For creating a web dashboard to visualize the Knowledge Graph and monitor the system performance.

Geospatial & Mapping:

- **Mapbox SDK:** Integrated for rendering interactive maps, showing the "Scenic Routes," and visualizing road elevation and safety zones.
- **Google Elevation API:** To fetch real time altitude and gradient data for the selected routes.

API Testing & Real time Automation:

- **Postman:** For testing and validating the accuracy of the GraphRAG reasoning responses and backend APIs.
- **node cron:** For scheduling automatic data fetching from official sources like the Disaster Management Centre (DMC) and Weather APIs.

3.2 System Architecture

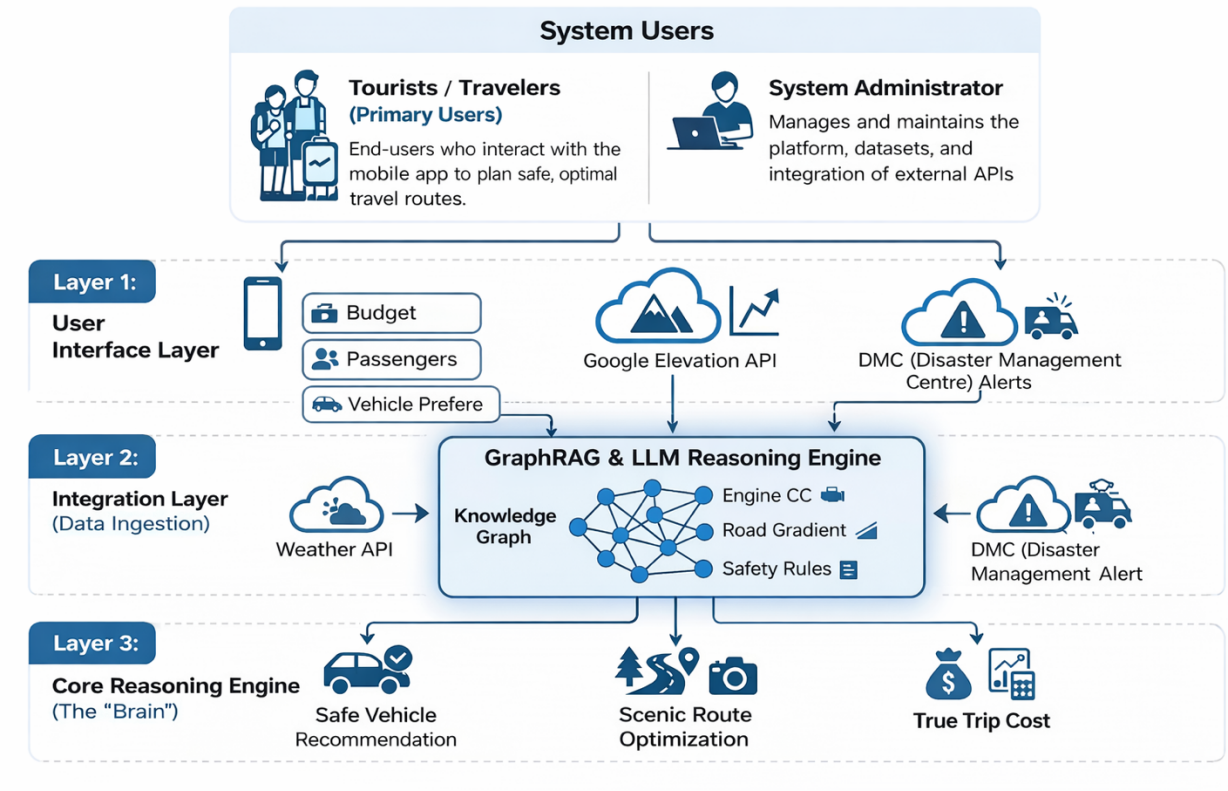


Figure 1: System Architecture

The **Intelligent Mobility and Vehicle Selection Engine** utilizes a multi tiered framework designed for real time terrain analysis and AI driven safety reasoning. The architecture is structured into four primary functional layers as shown in the diagram:

A. Layer 1: User Interface Layer This is the starting point of the system where the traveler interacts with the mobile application. The interface layer collects critical user inputs, including the **Travel Budget**, the **Number of Passengers**, and the **Initial Vehicle Preference**. These inputs are essential for the system to begin the context-aware recommendation process.

B. Layer 2: Integration Layer (Data Ingestion) The system ingests real-time data from three critical external streams to provide a high level of situational awareness:

- **Weather API:** Provides live environmental conditions like heavy rain or mist.
- **Google Elevation API:** Used to calculate the **Road Gradient** (steepness) for the selected route.
- **DMC (Disaster Management Centre) Alerts:** Fetches official hazard data such as landslide risks or floods specifically for the Sri Lankan context.

C. Layer 3: Core Reasoning Engine (The "Brain") This is the most important part of the architecture, where the **GraphRAG and LLM Reasoning Engine** operate.

- **Knowledge Graph Integration:** Unlike standard AI, this engine uses a structured Knowledge Graph that connects **Engine CC**, **Torque**, and **Mechanical Limits** with specific road conditions.
- **Safety Rules & Reasoning:** The "Brain" performs a context aware risk assessment. It cross references the vehicle's power against the road steepness and weather to generate a justification for why a vehicle is safe or unsafe.

D. Layer 4: Output Layer The final processed intelligence is pushed back to the traveler through three main outputs:

- **Safe Vehicle Recommendation:** Suggesting the most suitable vehicle based on safety and mechanics.
- **Scenic Route Optimization:** Recalculating the path to show beautiful POIs while avoiding risky areas.
- **True Trip Cost:** Providing an accurate financial estimation based on real time engine strain and distance.

3.3 Component Overview Diagram

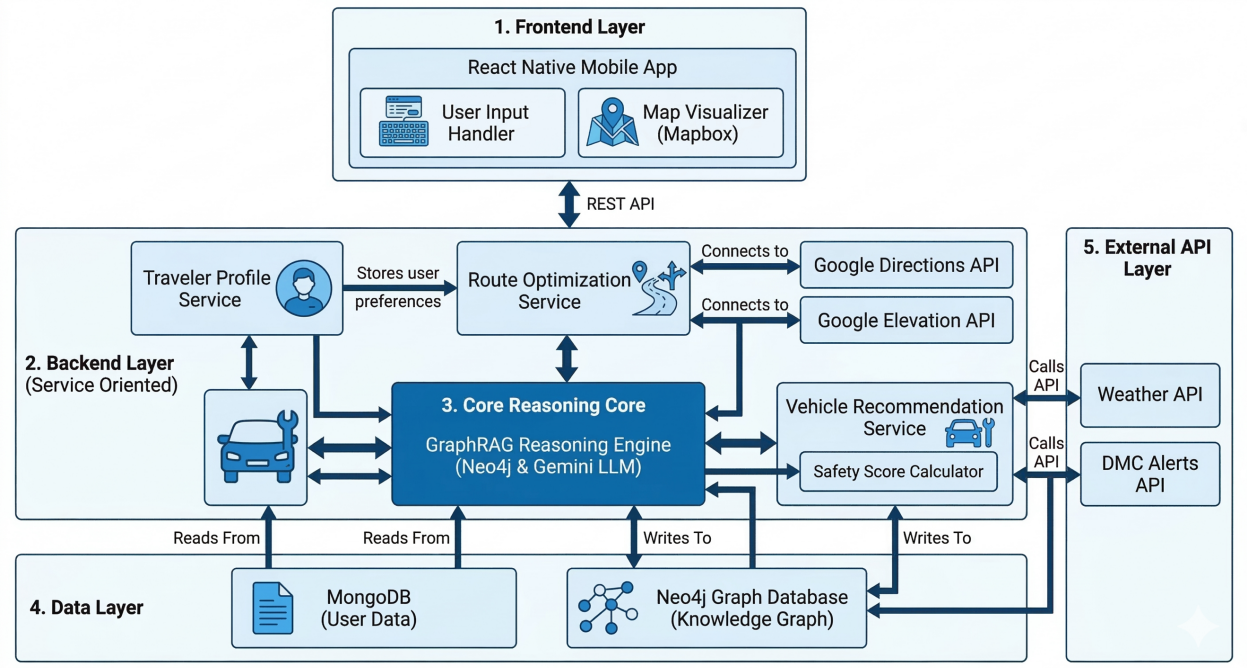


Figure 2: Component Overview Diagram

The component overview diagram provides a detailed technical view of the internal software modules. The system is built using a service oriented approach. The **React Native Mobile App** handles user interaction and displays the scenic route. The core functionality resides in the backend microservices. The **Route Optimization Service** calculates gradients using the Google Elevation API, while the **Vehicle Recommendation Service** runs the safety score logic. All services are orchestrated by the **GraphRAG Reasoning Engine**, which utilizes the **Neo4j Graph Database** to retrieve structured knowledge about vehicles and roads, combined with the **Gemini LLM** for contextual understanding. Data is persisted in **MongoDB**, and real time updates are fetched from external weather and disaster alert APIs.

3.4 Work Breakdown Structure

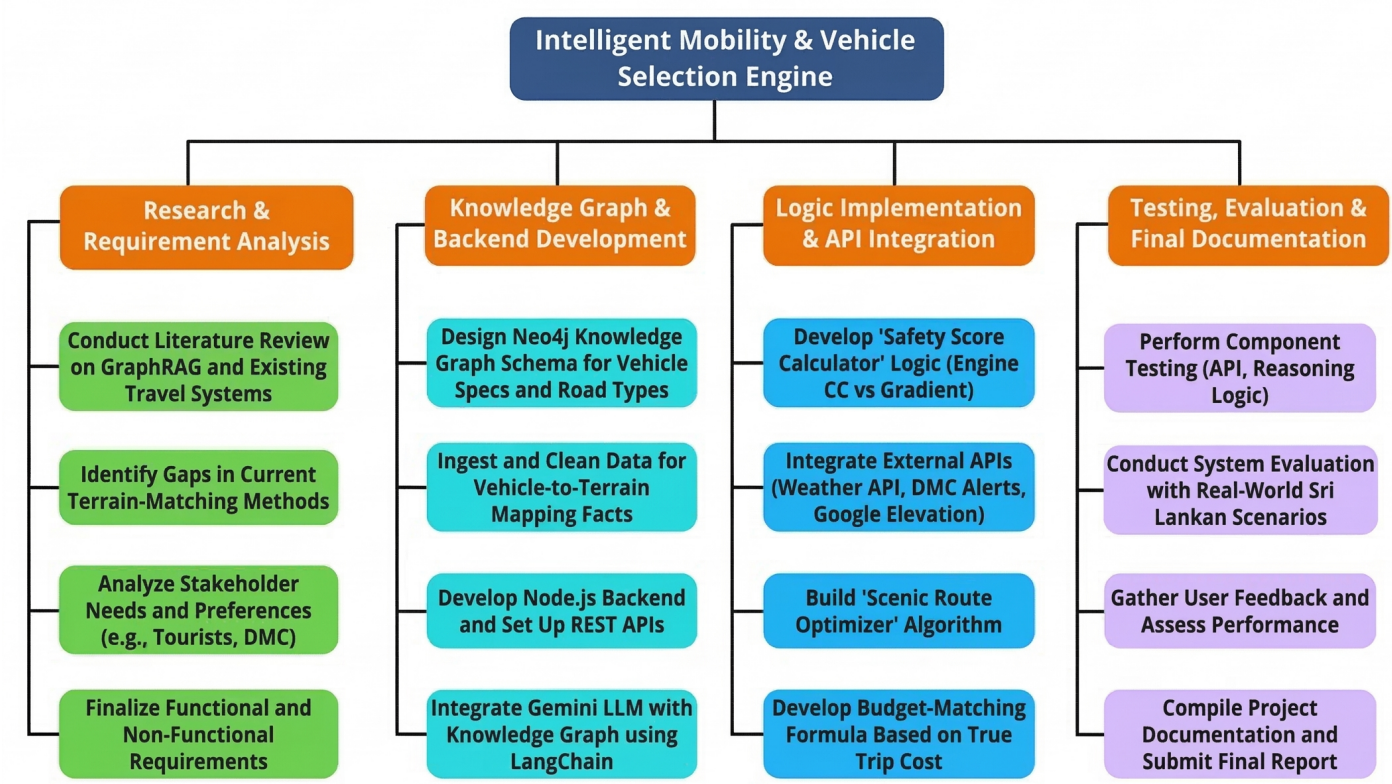


Figure 3: Work Breakdown Structure

3.5 Gantt Chart

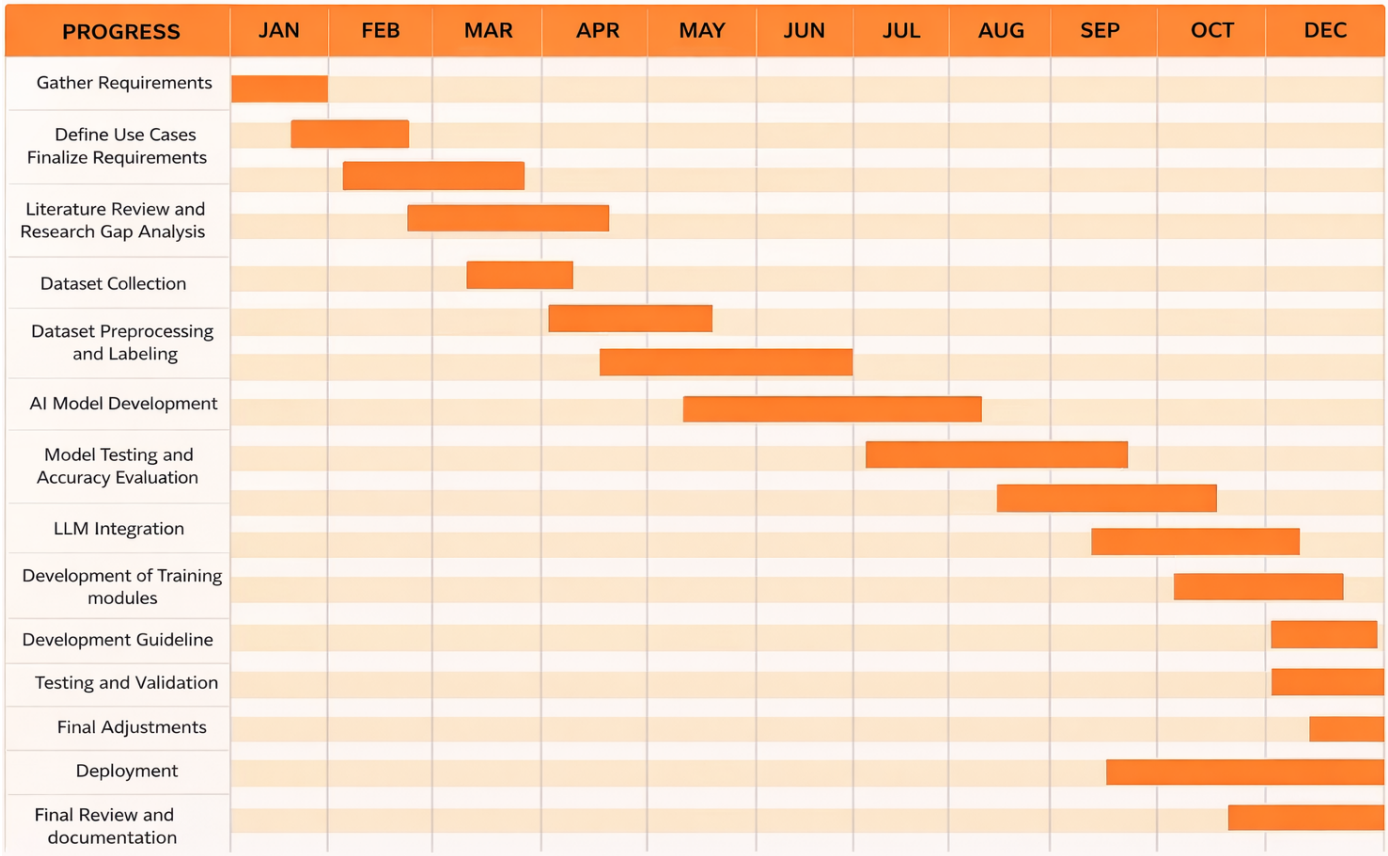


Figure 4: Gantt Chart

4. PROJECT REQUIREMENTS

This section explains the main things that the system must do and how it should perform to satisfy the travelers in Sri Lanka. I have divided these requirements into two parts: Functional and Non-functional.

4.1 Functional Requirements (FR)

These are the core features that the Intelligent Vehicle Selection and Route Optimization engine must provide:

- **User Input Processing:** The system must allow travelers to enter their starting point, destination, budget, number of passengers, and their preferred vehicle type.
- **Knowledge Retrieval (GraphRAG):** The system must fetch relevant data from the Knowledge Graph about road gradients and vehicle engine specifications (CC) based on the user's route.
- **Safety Score Calculation:** The engine must automatically calculate a safety score by comparing the vehicle's power with the road's steepness and current weather conditions.
- **Real time Alert Integration:** The system must fetch and show live alerts from the Disaster Management Centre (DMC) and Weather APIs to identify any risks like landslides on the path.
- **Vehicle Recommendation:** If the traveler's choice is not safe, the system must suggest a better vehicle alternative that is suitable for that specific terrain and weather.
- **Scenic Route Generation:** The system must provide an alternative "Scenic Route" that includes verified points of interest (POIs) while maintaining the safety standards.
- **True Trip Cost Estimation:** The system must calculate an accurate fuel cost by looking at the engine strain on mountain roads instead of just distance.

4.2 Non-Functional Requirements (NFR)

These are the quality standards the system must maintain:

- **Accuracy:** The vehicle recommendations and safety alerts must be very accurate because they are related to the traveler's life and safety.
- **Response Time (Latency):** The system should provide the route and vehicle suggestions within a few seconds (low latency) even when fetching real time API data.
- **Scalability:** The backend must handle many users at the same time, especially during the peak tourism seasons in Sri Lanka.
- **Explainability:** The AI should not just say "No," it must provide a simple reason why a vehicle is not recommended (e.g., "Engine capacity too low for this steep road").
- **Availability:** The system and its Knowledge Graph must be available 24/7 for travelers who are planning their trips at any time.

4.3 System Requirements

To develop and run this Intelligent Vehicle Selection and Route Optimization system, the following hardware and software requirements must be met:

A. Hardware Requirements

- **Development Workstation:** A computer with at least 8GB RAM (16GB recommended) and an i5 or higher processor to handle the Node.js backend and Neo4j graph database locally.
- **Mobile Device:** An Android or iOS smartphone for testing the React Native application in real world travel conditions.
- **Cloud Infrastructure:** A cloud server (like AWS or Google Cloud) to host the backend and the Knowledge Graph so travelers can access it anytime.
- **Internet Connectivity:** Stable high-speed internet is needed for the system to fetch real time weather and DMC data from external APIs.

B. Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux for the development environment.
- **Database Management System: Neo4j Desktop** for managing the Knowledge Graph and **MongoDB Compass**for viewing user data.
- **Integrated Development Environment (IDE): VS Code** for writing the backend and frontend code.
- **Runtime Environment: Node.js (LTS version)** to execute the backend logic and API services.
- **API Testing Tool: Postman** to check if the AI reasoning and data fetching work correctly.
- **Version Control: Git** and **GitHub** to manage the source code and track changes during the project.

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